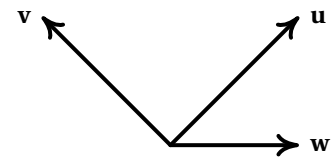
1. Consider the vectors $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ in the plane of this piece of paper. Circle the best answer. **(5 points)**



$\mathbf{w} =$ | $\mathbf{u}+\mathbf{v}$ | $\mathbf{u}-\mathbf{v}$ | $\frac{\mathbf{u}+\mathbf{v}}{2}$ | $\frac{\mathbf{u}-\mathbf{v}}{2}$ | $\mathbf{u}-\frac{\mathbf{v}}{2}$ | $\frac{\mathbf{v}-\mathbf{u}}{2}$ |

2. Consider the three points $A = (0,1,0)$, $B = (1,1,1)$, and $C = (0,2,1)$ in $\mathbb{R}^3$. For each part, circle the best answer. **(5 points each)**

(a) the projection of the vector $\overrightarrow{AC}$ onto $\overrightarrow{AB}$ is: | $\langle 0,2,2\rangle$ | $\left\langle \frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right\rangle$ | $\left\langle \frac{1}{2}, 0, \frac{1}{2} \right\rangle$ | $\left\langle \frac{1}{2}, \frac{1}{2}, 0 \right\rangle$ |

(b) The area of the triangle formed by these three points is: | 0 | $\sqrt{3}$ | $\frac{\sqrt{3}}{2}$ | 3 | $\frac{3}{2}$ |

(c) $\overrightarrow{AB} \cdot (\overrightarrow{AB} \times \overrightarrow{AC}) =$ | -3 | -2 | -1 | 0 | 1 | 2 | 3 |

3. Let θ be the angle between the vectors $\mathbf{u} = \langle 3,-4,0\rangle$ and $\mathbf{v} = \langle 2,1,2\rangle$. Circle the value of $\cos\theta$. **(5 points)**

| -1 | $-\frac{2}{15}$ | $\frac{1}{\sqrt{2}}$ | $-\frac{2}{\sqrt{15}}$ | 0 | $\frac{2}{\sqrt{15}}$ | $\frac{1}{\sqrt{2}}$ | $\frac{2}{15}$ | 1 |

Scratch Space

4. Let $\mathbf{r}(t) = \langle t+1, t^2, t^3+t \rangle$.

(a) Write a vector function $\mathbf{s}(t)$ representing the tangent line to the curve at $t = 0$. **(5 points)**

$\mathbf{s}(t) = \langle \qquad , \qquad , \qquad \rangle$

(b) Find the unit binormal vector $\mathbf{B}$ to the curve at $t = 0$. **(15 points)**

$\mathbf{B} = \langle \qquad , \qquad , \qquad \rangle$

(c) Choose the statement that best describes the unit binormal vector $\mathbf{B}$ for a general value of t. **(5 points)**

- [ ] The unit binormal vector $\mathbf{B}$ is parallel to the vector $\mathbf{r}(t)$.
- [ ] The unit binormal vector $\mathbf{B}$ is parallel to the vector $\mathbf{r}'(t)$.
- [ ] The unit binormal vector $\mathbf{B}$ is parallel to the vector $\mathbf{r}''(t)$.
- [ ] The unit binormal vector $\mathbf{B}$ is perpendicular to the osculating plane of the curve.
- [ ] The unit binormal vector $\mathbf{B}$ always points toward the center of curvature.

5. The vector functions $\mathbf{r}_1(t) = \langle t, 3+t, 3+t \rangle$ and $\mathbf{r}_2(t) = \langle 1+t, 4+t, -t \rangle$ represent two lines.

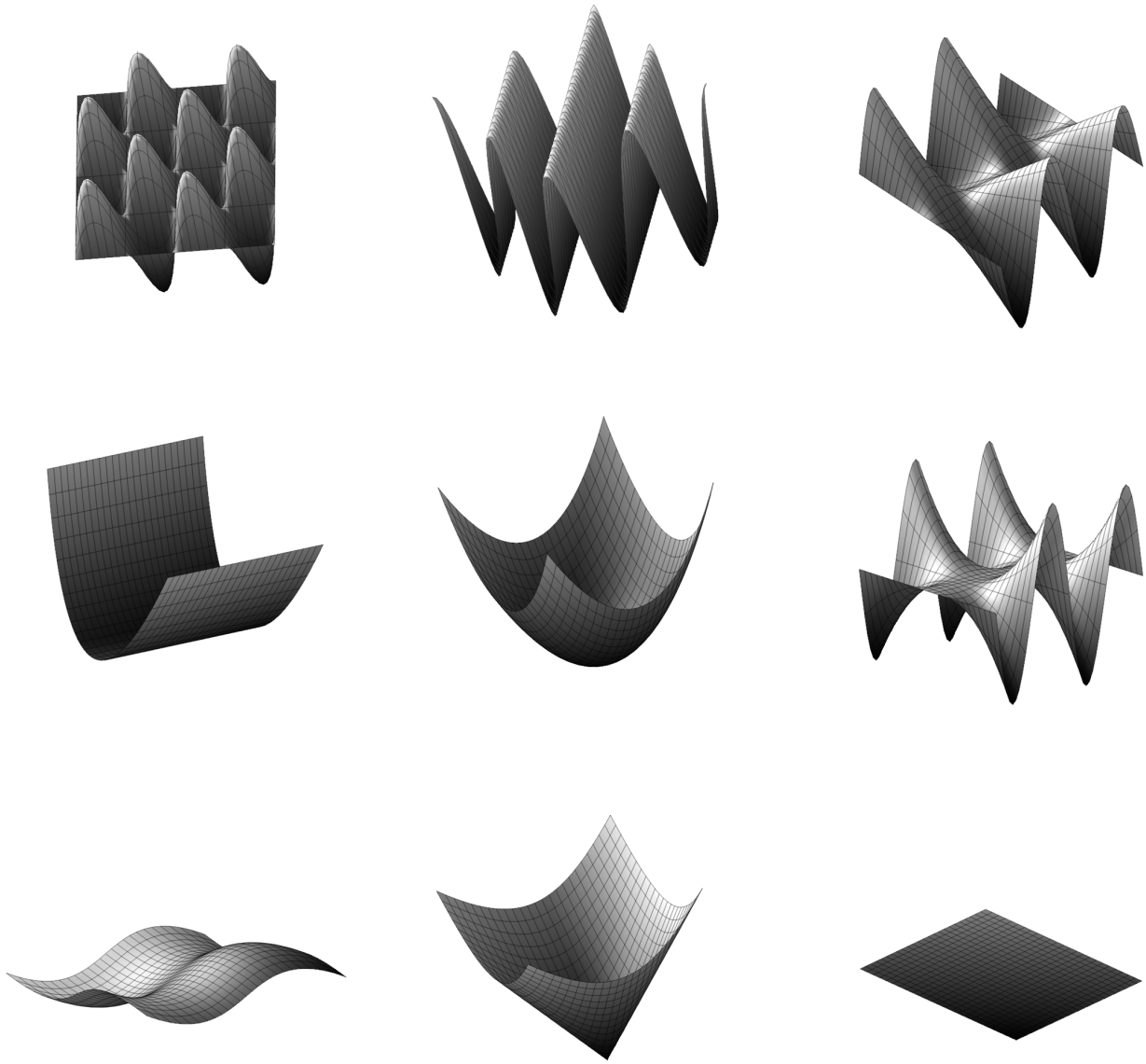
(a) Find a point P that lies on both lines. **(5 points)**

$P = \left(\quad , \quad , \quad \right)$

(b) Find an equation of the plane containing both lines. **(15 points)**

Equation: $\boxed{}\, x + \boxed{}\, y + \boxed{}\, z = \boxed{}$

6. Circle the graph that represents the function $f(x, y) = y \sin x$. **(10 points)**



Scratch Space

7. Consider the function

$$f(x, y) = \frac{(x+y)y}{x^2+y^2}$$

defined when $(x, y) \neq (0, 0)$.

Evaluate

$$\lim_{(x,y)\to(0,0)} f(x, y)$$

or explain why it does not exist. If the limit does not exist, write "DNE" in the box provided. **(20 points)**

Limit = $\boxed{}$